

Final Project Survival

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```
library(survival)
library(gtsummary)
```

```
## Warning: package 'gtsummary' was built under R version 4.5.2
```

```
library(cmprsk)
```

```
## Warning: package 'cmprsk' was built under R version 4.5.3
```

```
set.seed(120401)
```

Step 2: Summary of variables

```
# Defining PBC (Primary Biliary Cholangitis)
```

```
myid = sample(pbc$id, 400)
```

```
mypbc = subset(pbc, id %in% myid)
```

```
# Variables
```

```
vars <- mypbc[, c("age", "bili", "albumin", "platelet", "sex",  
                  "edema", "ascites", "stage", "trt", "status", "time")]
```

```
# summary table
```

```
table1 <- tbl_summary(  
  vars,  
  statistic = list(  
    all_continuous() ~ "{mean} ({sd})",  
    all_categorical() ~ "{n} ({p}%)"  
  ),  
  missing_text = "Missing"  
)
```

```
table1
```

```
print(table1)
```

```
## <div id="emkmvnmsem" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;
##   <style>#emkmvnmsem table {
##     font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Se
##   -webkit-font-smoothing: antialiased;
##   -moz-osx-font-smoothing: grayscale;
## }
##
## #emkmvnmsem thead, #emkmvnmsem tbody, #emkmvnmsem tfoot, #emkmvnmsem tr, #emkmvnmsem td, #emkmvnmsem
##   border-style: none;
## }
```

Characteristic	N = 400 ¹
age	51 (11)
bili	3.2 (4.4)
albumin	3.50 (0.42)
platelet	256 (97)
Missing	10
sex	
m	43 (11%)
f	357 (89%)
edema	
0	340 (85%)
0.5	41 (10%)
1	19 (4.8%)
ascites	23 (7.6%)
Missing	99
stage	
1	20 (5.1%)
2	87 (22%)
3	148 (38%)
4	139 (35%)
Missing	6
trt	
1	154 (51%)
2	147 (49%)
Missing	99
status	
0	219 (55%)
1	24 (6.0%)
2	157 (39%)
time	1,914 (1,098)

¹Mean (SD); n (%)

```
##
## #emkmvnmsem p {
##   margin: 0;
##   padding: 0;
## }
##
## #emkmvnmsem .gt_table {
##   display: table;
##   border-collapse: collapse;
##   line-height: normal;
##   margin-left: auto;
##   margin-right: auto;
##   color: #333333;
##   font-size: 16px;
```

```

## font-weight: normal;
## font-style: normal;
## background-color: #FFFFFF;
## width: auto;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #A8A8A8;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #A8A8A8;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_caption {
## padding-top: 4px;
## padding-bottom: 4px;
## }
##
## #emkmvnmsem .gt_title {
## color: #333333;
## font-size: 125%;
## font-weight: initial;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## border-bottom-color: #FFFFFF;
## border-bottom-width: 0;
## }
##
## #emkmvnmsem .gt_subtitle {
## color: #333333;
## font-size: 85%;
## font-weight: initial;
## padding-top: 3px;
## padding-bottom: 5px;
## padding-left: 5px;
## padding-right: 5px;
## border-top-color: #FFFFFF;
## border-top-width: 0;
## }
##
## #emkmvnmsem .gt_heading {
## background-color: #FFFFFF;
## text-align: center;
## border-bottom-color: #FFFFFF;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;

```

```

## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_bottom_border {
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_col_headings {
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_col_heading {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 6px;
## padding-left: 5px;
## padding-right: 5px;
## overflow-x: hidden;
## }
##
## #emkmvnmsem .gt_column_spanner_outer {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## padding-top: 0;
## padding-bottom: 0;

```

```

## padding-left: 4px;
## padding-right: 4px;
## }
##
## #emkmvnmsem .gt_column_spanner_outer:first-child {
## padding-left: 0;
## }
##
## #emkmvnmsem .gt_column_spanner_outer:last-child {
## padding-right: 0;
## }
##
## #emkmvnmsem .gt_column_spanner {
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 5px;
## overflow-x: hidden;
## display: inline-block;
## width: 100%;
## }
##
## #emkmvnmsem .gt_spanner_row {
## border-bottom-style: hidden;
## }
##
## #emkmvnmsem .gt_group_heading {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: initial;
## text-transform: inherit;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: middle;
## text-align: left;
## }
##

```

```

## #emkmvnmsem .gt_empty_group_heading {
##   padding: 0.5px;
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
##   vertical-align: middle;
## }
##
## #emkmvnmsem .gt_from_md > :first-child {
##   margin-top: 0;
## }
##
## #emkmvnmsem .gt_from_md > :last-child {
##   margin-bottom: 0;
## }
##
## #emkmvnmsem .gt_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   margin: 10px;
##   border-top-style: solid;
##   border-top-width: 1px;
##   border-top-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
##   vertical-align: middle;
##   overflow-x: hidden;
## }
##
## #emkmvnmsem .gt_stub {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
## }

```

```

##
## #emkmvnmsem .gt_stub_row_group {
##   color: #333333;
##   background-color: #FFFFFF;
##   font-size: 100%;
##   font-weight: initial;
##   text-transform: inherit;
##   border-right-style: solid;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   padding-left: 5px;
##   padding-right: 5px;
##   vertical-align: top;
## }
##
## #emkmvnmsem .gt_row_group_first td {
##   border-top-width: 2px;
## }
##
## #emkmvnmsem .gt_row_group_first th {
##   border-top-width: 2px;
## }
##
## #emkmvnmsem .gt_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #emkmvnmsem .gt_first_summary_row {
##   border-top-style: solid;
##   border-top-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_first_summary_row.thick {
##   border-top-width: 2px;
## }
##
## #emkmvnmsem .gt_last_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_grand_summary_row {
##   color: #333333;

```

```

## background-color: #FFFFFF;
## text-transform: inherit;
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## }
##
## #emkmvnmsem .gt_first_grand_summary_row {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## border-top-style: double;
## border-top-width: 6px;
## border-top-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_last_grand_summary_row_top {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## border-bottom-style: double;
## border-bottom-width: 6px;
## border-bottom-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_stripped {
## background-color: rgba(128, 128, 128, 0.05);
## }
##
## #emkmvnmsem .gt_table_body {
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_footnotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## }

```



```

##
## #emkmvnmsem .gt_footnote {
##     margin: 0px;
##     font-size: 90%;
##     padding-top: 4px;
##     padding-bottom: 4px;
##     padding-left: 5px;
##     padding-right: 5px;
## }
##
## #emkmvnmsem .gt_sourcenotes {
##     color: #333333;
##     background-color: #FFFFFF;
##     border-bottom-style: none;
##     border-bottom-width: 2px;
##     border-bottom-color: #D3D3D3;
##     border-left-style: none;
##     border-left-width: 2px;
##     border-left-color: #D3D3D3;
##     border-right-style: none;
##     border-right-width: 2px;
##     border-right-color: #D3D3D3;
## }
##
## #emkmvnmsem .gt_sourcenote {
##     font-size: 90%;
##     padding-top: 4px;
##     padding-bottom: 4px;
##     padding-left: 5px;
##     padding-right: 5px;
## }
##
## #emkmvnmsem .gt_left {
##     text-align: left;
## }
##
## #emkmvnmsem .gt_center {
##     text-align: center;
## }
##
## #emkmvnmsem .gt_right {
##     text-align: right;
##     font-variant-numeric: tabular-nums;
## }
##
## #emkmvnmsem .gt_font_normal {
##     font-weight: normal;
## }
##
## #emkmvnmsem .gt_font_bold {
##     font-weight: bold;
## }
##
## #emkmvnmsem .gt_font_italic {

```

```

##   font-style: italic;
## }
##
## #emkmvnmsem .gt_super {
##   font-size: 65%;
## }
##
## #emkmvnmsem .gt_footnote_marks {
##   font-size: 75%;
##   vertical-align: 0.4em;
##   position: initial;
## }
##
## #emkmvnmsem .gt_asterisk {
##   font-size: 100%;
##   vertical-align: 0;
## }
##
## #emkmvnmsem .gt_indent_1 {
##   text-indent: 5px;
## }
##
## #emkmvnmsem .gt_indent_2 {
##   text-indent: 10px;
## }
##
## #emkmvnmsem .gt_indent_3 {
##   text-indent: 15px;
## }
##
## #emkmvnmsem .gt_indent_4 {
##   text-indent: 20px;
## }
##
## #emkmvnmsem .gt_indent_5 {
##   text-indent: 25px;
## }
##
## #emkmvnmsem .katex-display {
##   display: inline-flex !important;
##   margin-bottom: 0.75em !important;
## }
##
## #emkmvnmsem div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-g
##   height: 0px !important;
## }
## </style>
##   <table class="gt_table" data-quarto-disable-processing="false" data-quarto-bootstrap="false">
##     <thead>
##       <tr class="gt_col_headings">
##         <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##         <th class="gt_col_heading gt_columns_bottom_border gt_center" rowspan="1" colspan="1" scope="c
##       </tr>
##     </thead>

```

```

## <tbody class="gt_table_body">
## <tr><td headers="label" class="gt_row gt_left">age</td>
## <td headers="stat_0" class="gt_row gt_center">51 (11)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">bili</td>
## <td headers="stat_0" class="gt_row gt_center">3.2 (4.4)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">albumin</td>
## <td headers="stat_0" class="gt_row gt_center">3.50 (0.42)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">platelet</td>
## <td headers="stat_0" class="gt_row gt_center">256 (97)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">Missing</td>
## <td headers="stat_0" class="gt_row gt_center">10</td></tr>
## <tr><td headers="label" class="gt_row gt_left">sex</td>
## <td headers="stat_0" class="gt_row gt_center"><br /></td></tr>
## <tr><td headers="label" class="gt_row gt_left">m</td>
## <td headers="stat_0" class="gt_row gt_center">43 (11%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">f</td>
## <td headers="stat_0" class="gt_row gt_center">357 (89%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">edema</td>
## <td headers="stat_0" class="gt_row gt_center"><br /></td></tr>
## <tr><td headers="label" class="gt_row gt_left">0</td>
## <td headers="stat_0" class="gt_row gt_center">340 (85%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">0.5</td>
## <td headers="stat_0" class="gt_row gt_center">41 (10%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">1</td>
## <td headers="stat_0" class="gt_row gt_center">19 (4.8%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">ascites</td>
## <td headers="stat_0" class="gt_row gt_center">23 (7.6%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">Missing</td>
## <td headers="stat_0" class="gt_row gt_center">99</td></tr>
## <tr><td headers="label" class="gt_row gt_left">stage</td>
## <td headers="stat_0" class="gt_row gt_center"><br /></td></tr>
## <tr><td headers="label" class="gt_row gt_left">1</td>
## <td headers="stat_0" class="gt_row gt_center">20 (5.1%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">2</td>
## <td headers="stat_0" class="gt_row gt_center">87 (22%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">3</td>
## <td headers="stat_0" class="gt_row gt_center">148 (38%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">4</td>
## <td headers="stat_0" class="gt_row gt_center">139 (35%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">Missing</td>
## <td headers="stat_0" class="gt_row gt_center">6</td></tr>
## <tr><td headers="label" class="gt_row gt_left">trt</td>
## <td headers="stat_0" class="gt_row gt_center"><br /></td></tr>
## <tr><td headers="label" class="gt_row gt_left">1</td>
## <td headers="stat_0" class="gt_row gt_center">154 (51%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">2</td>
## <td headers="stat_0" class="gt_row gt_center">147 (49%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">Missing</td>
## <td headers="stat_0" class="gt_row gt_center">99</td></tr>
## <tr><td headers="label" class="gt_row gt_left">status</td>
## <td headers="stat_0" class="gt_row gt_center"><br /></td></tr>
## <tr><td headers="label" class="gt_row gt_left">0</td>
## <td headers="stat_0" class="gt_row gt_center">219 (55%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">1</td>

```

```
## <td headers="stat_0" class="gt_row gt_center">24 (6.0%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left"> 2</td>
## <td headers="stat_0" class="gt_row gt_center">157 (39%)</td></tr>
## <tr><td headers="label" class="gt_row gt_left">time</td>
## <td headers="stat_0" class="gt_row gt_center">1,914 (1,098)</td></tr>
## </tbody>
## <tfoot>
## <tr class="gt_footnotes">
## <td class="gt_footnote" colspan="2"><span class="gt_footnote_marks" style="white-space:nowrap;
## </tr>
## </tfoot>
## </table>
## </div>
```

Step 3: Kaplan-Meier Survival Curves

```
# in the default status: 0 = censored, 1 = transplant, 2 = dead
# patientsw/transplant = censored (time of their transplant)
mypbc$event_death <- ifelse(mypbc$status == 2, 1, 0) #KM curve for time to death
#patientsw/death = censored
mypbc$event_transplant <- ifelse(mypbc$status == 1, 1, 0) #KM curve for time to liver transplant

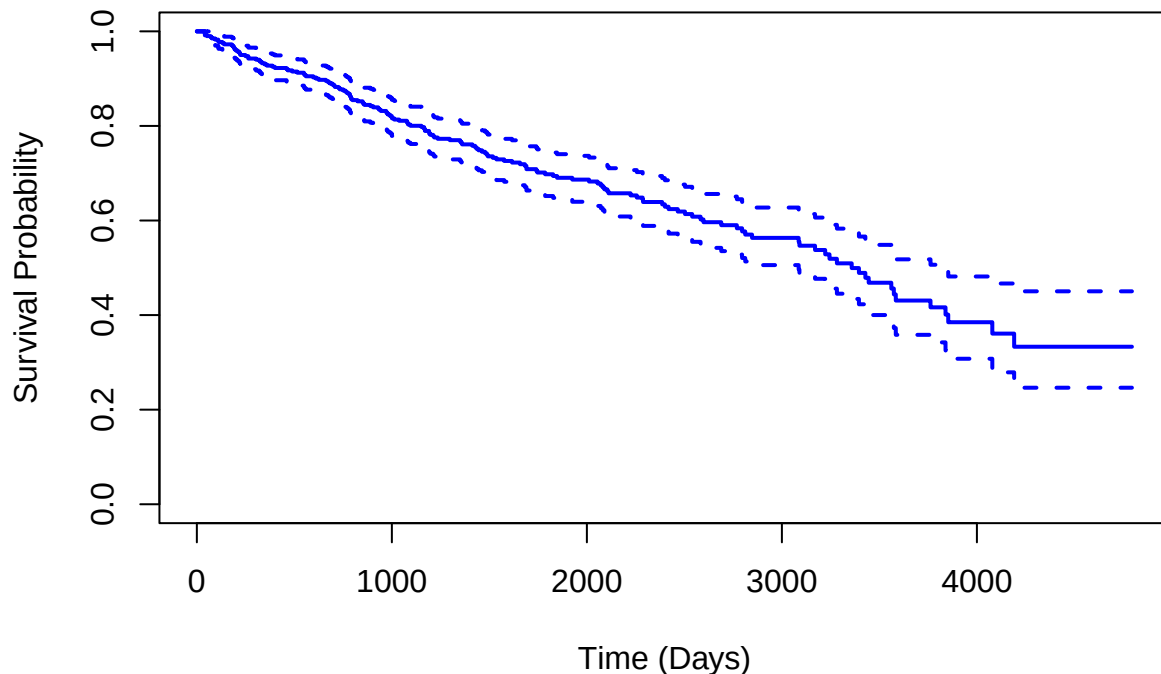
#KM Curve for event_death
km_death <- survfit(Surv(time, event_death) ~ 1, data = mypbc)
print(km_death, print.rmean = TRUE)
```

```
## Call: survfit(formula = Surv(time, event_death) ~ 1, data = mypbc)
##
##          n events rmean* se(rmean) median 0.95LCL 0.95UCL
## [1,] 400      157   3016         101   3358     3086     3839
##      * restricted mean with upper limit = 4795
```

```
plot(km_death,
     main = "Kaplan-Meier Curve for Time to Death",
     xlab = "Time (Days)",
     ylab = "Survival Probability",
     col = "blue",
     lwd = 2,
     mark.table = TRUE)
```

```
## Warning in plot.window(...): "mark.table" is not a graphical parameter
## Warning in plot.xy(xy, type, ...): "mark.table" is not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "mark.table" is
## not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "mark.table" is
## not a graphical parameter
## Warning in box(...): "mark.table" is not a graphical parameter
## Warning in title(...): "mark.table" is not a graphical parameter
```

Kaplan–Meier Curve for Time to Death



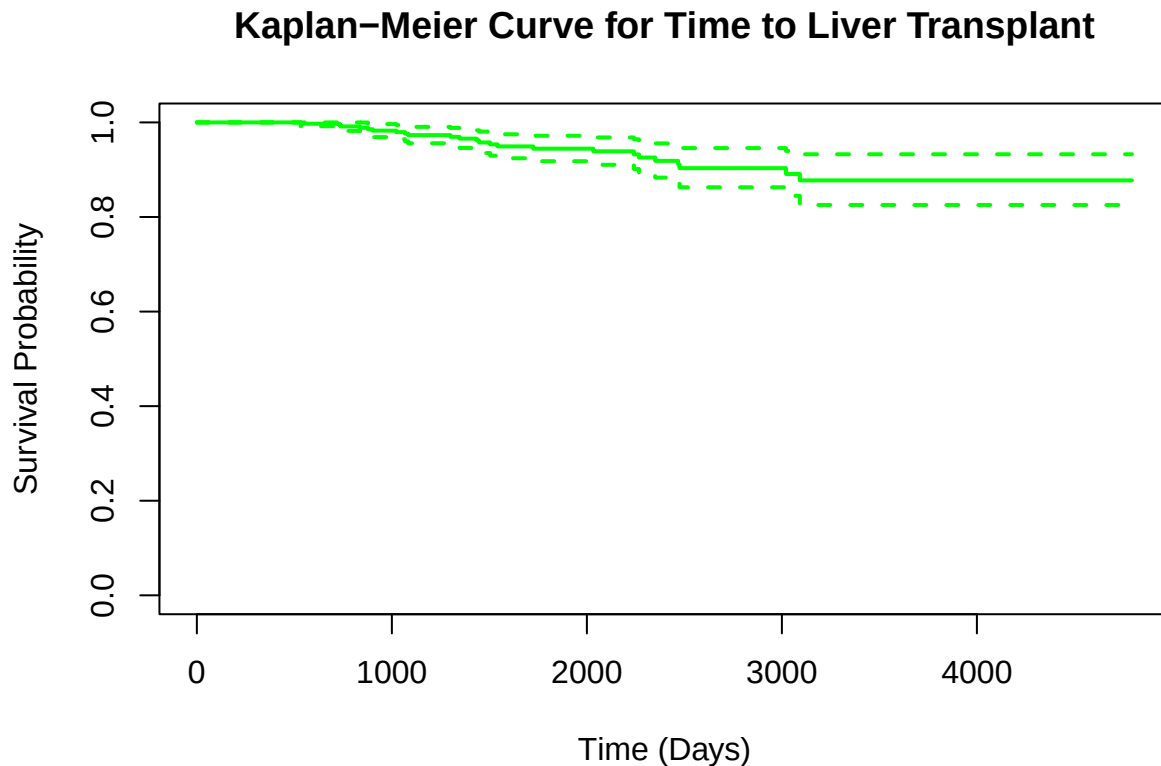
```
#KM Curve for event_transplant
km_trans <- survfit(Surv(time, event_transplant) ~ 1, data = mypbc)
print(km_trans, print.rmean = TRUE)

## Call: survfit(formula = Surv(time, event_transplant) ~ 1, data = mypbc)
##
##          n events rmean* se(rmean) median 0.95LCL 0.95UCL
## [1,] 400      24  4448          69      NA      NA      NA
##      * restricted mean with upper limit =  4795

plot(km_trans,
      main = "Kaplan-Meier Curve for Time to Liver Transplant",
      xlab = "Time (Days)",
      ylab = "Survival Probability",
      col = "green",
      lwd = 2,
      mark.table = TRUE)

## Warning in plot.window(...): "mark.table" is not a graphical parameter
## Warning in plot.xy(xy, type, ...): "mark.table" is not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "mark.table" is
## not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "mark.table" is
## not a graphical parameter
## Warning in box(...): "mark.table" is not a graphical parameter
```

```
## Warning in title(...): "mark.table" is not a graphical parameter
```



Step 4: Cox Proportional Hazards Model

```
# Make sure to have status coming out
vars_for_model <- c("time", "event_death", "age", "bili", "albumin",
                   "platelet", "sex", "edema", "ascites", "stage", "trt", "status")

mypbc_complete <- na.omit(mypbc[,vars_for_model])
# Cox proportional Hazards Model (Time to Death)

##### Double-Check on these variables bili, edema and stage #####
# Are they suppose to be all log or log and strata
# Strata - stage and edema are categorical variables (Ask Yohanna)
# log transformation - requires continous variables
cox_log <- coxph(Surv(time, event_death) ~ age +
                 log(bili) + albumin + sex + strata(edema) +
                 strata(stage), data = mypbc_complete)

summary(cox_log)

## Call:
## coxph(formula = Surv(time, event_death) ~ age + log(bili) + albumin +
##       sex + strata(edema) + strata(stage), data = mypbc_complete)
##
```

```

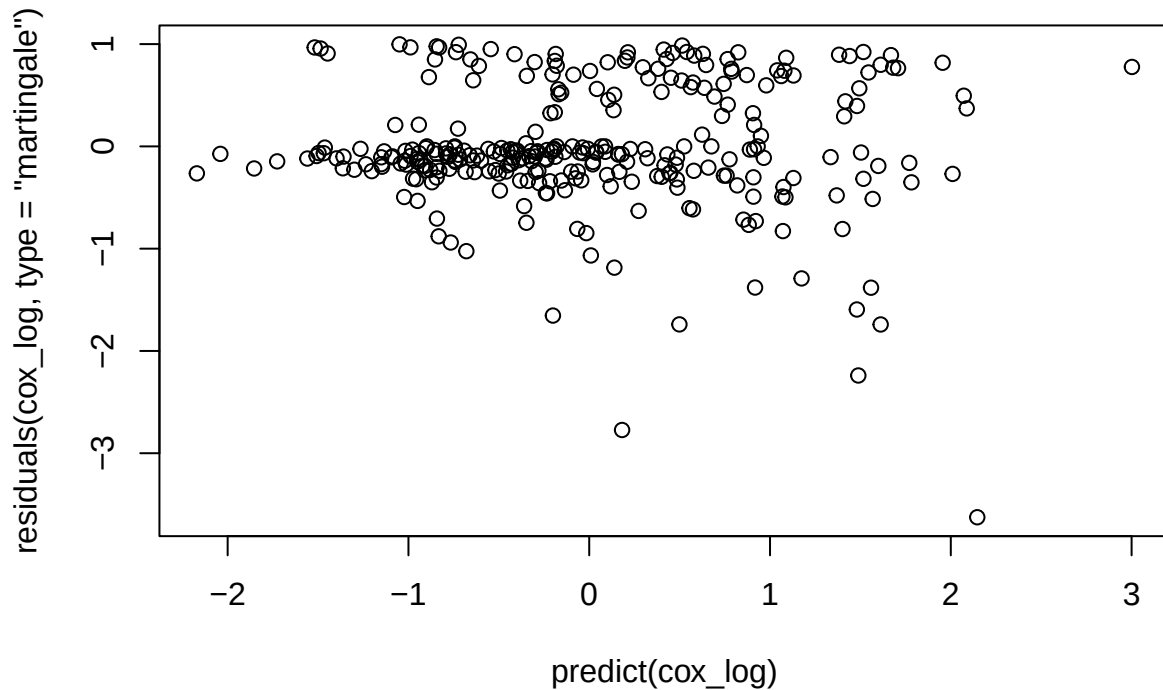
## n= 297, number of events= 120
##
##          coef exp(coef) se(coef)      z Pr(>|z|)
## age      0.027972  1.028367  0.009916  2.821  0.00479 **
## log(bili) 0.842055  2.321133  0.112145  7.509 5.97e-14 ***
## albumin  -0.636906  0.528926  0.262101 -2.430  0.01510 *
## sexf      -0.310358  0.733184  0.268191 -1.157  0.24718
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##          exp(coef) exp(-coef) lower .95 upper .95
## age      1.0284      0.9724      1.0086      1.0485
## log(bili) 2.3211      0.4308      1.8631      2.8917
## albumin   0.5289      1.8906      0.3164      0.8841
## sexf       0.7332      1.3639      0.4334      1.2402
##
## Concordance= 0.738 (se = 0.034 )
## Likelihood ratio test= 83 on 4 df,  p=<2e-16
## Wald test              = 76.84 on 4 df,  p=8e-16
## Score (logrank) test = 86.29 on 4 df,  p=<2e-16

print(cox.zph(cox_log))

##          chisq df      p
## age      1.88e-06 1 0.999
## log(bili) 3.35e+00 1 0.067
## albumin   8.36e-01 1 0.361
## sex       3.03e-02 1 0.862
## GLOBAL    4.19e+00 4 0.381

# Wanna double-check
plot(predict(cox_log), residuals(cox_log, type = "martingale"))

```



```
# WeIBull AFT model
wei_aft <- survreg(Surv(time, event_death) ~ age + log(bili) + albumin + factor(edema) + factor(stage),
  data = mypbc_complete, dist = "weibull")
wei_final <- step(wei_aft, direction = "both", trace = 0)
summary(wei_final)
```

```
##
## Call:
## survreg(formula = Surv(time, event_death) ~ age + log(bili) +
##   albumin + factor(edema) + factor(stage), data = mypbc_complete,
##   dist = "weibull")
##
```

	Value	Std. Error	z	p
## (Intercept)	8.79927	0.88465	9.95	< 2e-16
## age	-0.01834	0.00538	-3.41	0.00066
## log(bili)	-0.50477	0.06048	-8.35	< 2e-16
## albumin	0.44155	0.14966	2.95	0.00318
## factor(edema)0.5	-0.14126	0.17152	-0.82	0.41017
## factor(edema)1	-0.74524	0.17750	-4.20	2.7e-05
## factor(stage)2	-0.56493	0.62684	-0.90	0.36747
## factor(stage)3	-0.78129	0.61549	-1.27	0.20430
## factor(stage)4	-0.99708	0.61929	-1.61	0.10739
## Log(scale)	-0.50599	0.07238	-6.99	2.7e-12

```
##
## Scale= 0.603
##
## Weibull distribution
```



```

## Loglik(model)= -1044.4   Loglik(intercept only)= -1139
##  Chisq= 189.19 on 8 degrees of freedom, p= 1.2e-36
## Number of Newton-Raphson Iterations: 7
## n= 297

#Log-normal AFT Model
ln_full <- survreg(Surv(time, event_death) ~ age + log(bili) + albumin + sex + factor(edema) + factor(s
                        data = mypbc_complete, dist = "lognormal")

ln_final <- step(ln_full, direction = "both", trace = 0)
summary(ln_final)

##
## Call:
## survreg(formula = Surv(time, event_death) ~ age + log(bili) +
##      albumin + factor(edema) + factor(stage), data = mypbc_complete,
##      dist = "lognormal")
##              Value Std. Error      z      p
## (Intercept)    9.52522    0.95514  9.97 < 2e-16
## age           -0.02173    0.00644 -3.37 0.00074
## log(bili)      -0.54275    0.07078 -7.67 1.7e-14
## albumin        0.27583    0.17496  1.58 0.11491
## factor(edema)0.5 -0.31935    0.20687 -1.54 0.12266
## factor(edema)1   -1.08206    0.24002 -4.51 6.5e-06
## factor(stage)2   -0.61544    0.56725 -1.08 0.27794
## factor(stage)3   -0.90050    0.55380 -1.63 0.10394
## factor(stage)4   -1.24369    0.55812 -2.23 0.02586
## Log(scale)      -0.14715    0.06570 -2.24 0.02511
##
## Scale= 0.863
##
## Log Normal distribution
## Loglik(model)= -1045.6   Loglik(intercept only)= -1143.8
##  Chisq= 196.42 on 8 degrees of freedom, p= 3.6e-38
## Number of Newton-Raphson Iterations: 6
## n= 297

AIC(wei_final, ln_final)

##              df      AIC
## wei_final  10 2108.779
## ln_final   10 2111.257

Cumulative Incidence Functions

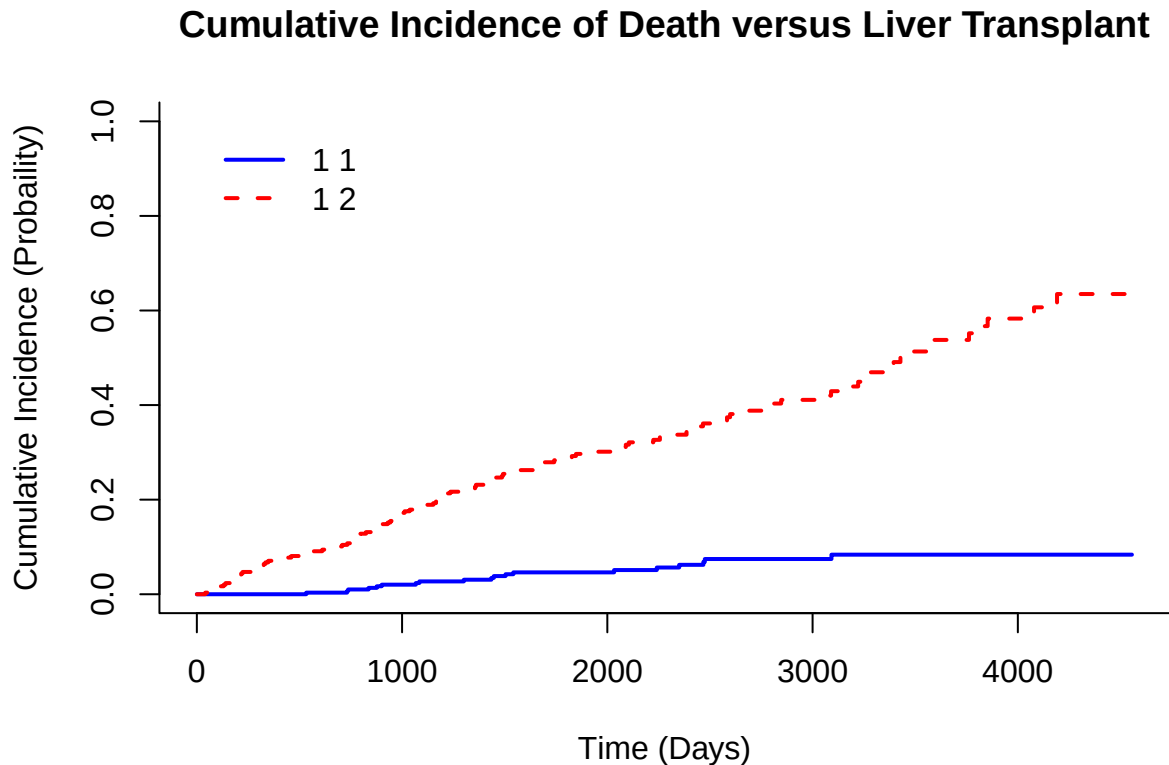
#CIF for all competing events with status column
cif <- cuminc(ftime = mypbc_complete$time, fstatus = mypbc_complete$status)
print(cif)

## Estimates and Variances:
## $est
##           1000           2000           3000           4000
## 1 1 0.02024132 0.04620904 0.0744674 0.08365493
## 1 2 0.17202521 0.30139517 0.4110308 0.58287865
##
## $var
##           1000           2000           3000           4000

```

```
## 1 1 6.716444e-05 0.0001582287 0.0003058708 0.0003840223
## 1 2 4.823787e-04 0.0007687083 0.0011345329 0.0020448303
```

```
plot(cif,
     xlab = "Time (Days)",
     ylab = "Cumulative Incidence (Probability)",
     main = "Cumulative Incidence of Death versus Liver Transplant",
     col = c("blue", "red"),
     lwd = 2)
```



Fine-Gray Subdistribution Hazard Model

```
matrix_cov <- model.matrix(~ age + log(bili) + albumin + factor(sex) + factor(edema) + factor(stage),
                           data = mypbc_complete)[,-1]

# Fine-Gray Model
# failcode = 2 death, cencode = 0 censored
fg_model <- crr(ftime = mypbc_complete$time,
               fstatus = mypbc_complete$status,
               cov1 = matrix_cov,
               failcode = 2,
               cencode = 0)

summary(fg_model)
```

```
## Competing Risks Regression
##
## Call:
## crr(ftime = mypbc_complete$time, fstatus = mypbc_complete$status,
```

```

##      cov1 = matrix_cov, failcode = 2, cencode = 0)
##
##               coef exp(coef) se(coef)      z p-value
## age           0.0345    1.035  0.0102  3.40 6.7e-04
## log(bili)      0.8175    2.265  0.1023  7.99 1.3e-15
## albumin       -0.6917    0.501  0.2464 -2.81 5.0e-03
## factor(sex)f   -0.3005    0.740  0.2916 -1.03 3.0e-01
## factor(edema)0.5 0.3044    1.356  0.2872  1.06 2.9e-01
## factor(edema)1  1.2612    3.530  0.3540  3.56 3.7e-04
## factor(stage)2  0.8943    2.446  0.7944  1.13 2.6e-01
## factor(stage)3  1.2446    3.472  0.7718  1.61 1.1e-01
## factor(stage)4  1.5530    4.725  0.7791  1.99 4.6e-02
##
##               exp(coef) exp(-coef)  2.5% 97.5%
## age           1.035    0.966 1.015 1.056
## log(bili)      2.265    0.442 1.853 2.768
## albumin        0.501    1.997 0.309 0.812
## factor(sex)f   0.740    1.351 0.418 1.311
## factor(edema)0.5 1.356    0.738 0.772 2.380
## factor(edema)1  3.530    0.283 1.764 7.064
## factor(stage)2  2.446    0.409 0.515 11.604
## factor(stage)3  3.472    0.288 0.765 15.756
## factor(stage)4  4.725    0.212 1.026 21.758
##
## Num. cases = 297
## Pseudo Log-likelihood = -524
## Pseudo likelihood ratio test = 180 on 9 df,

```